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RESEARCH ARTICLE

Phenological growth stages of quinoa (*Chenopodium quinoa*) based on the BBCH scale

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Abstract

Quinoa (*Chenopodium quinoa*) is a pseudocereal native from the Andean region of South America that has increased in importance worldwide. Quinoa is now considered an alternative to traditional crops in a climate change scenario, considering its ability to adapt to marginal soils, droughts and frosts. Despite the interesting agronomic and nutritional features of this crop, research into quinoa is characterised by individual attempts to define its phenological stages without an international consensus. A unique criterion to quantify the phenology of quinoa could become a useful tool for researchers and plant breeders in future work by standardising this information for international cooperation. In this article, a proposed scale of the phenological growth stages of quinoa based on the BBCH coding system (Biologische Bundesanstalt Bundessortenamt und CHemische Industrie) was developed. Growth stages were described utilising the decimal code of the BBCH system, and figures were included for the most representative stages.

Introduction

Quinoa (*Chenopodium quinoa* Willd., Amaranthaceae) is an herbaceous annual plant from the Andean region of South America that was cultivated by indigenous communities across Bolivia, Chile, Ecuador and Peru between 5000 and 750 B.C. (Ruas *et al.*, 1999). Quinoa cultivation expanded northward and southward, from 2°N in Colombia to 47°S in Chile (Bazile *et al.*, 2014; Zurita-Silva *et al.*, 2014), developing a large morphological and phenotypical plasticity among genotypes (Fuentes *et al.*, 2009; Fuentes & Bhargava, 2011). According to Zurita-Silva *et al.* (2014) and Graf *et al.* (2015) the worldwide interest in quinoa as a pseudocereal has grown extensively, thanks to its nutritional qualities and its ability to adapt to marginal soils with high salinity contents (Eisa *et al.*, 2012), dry environments prone to frosts (Jacobsen *et al.*, 2007) and scarce water resources (García *et al.*, 2003). This adaptability has allowed quinoa to be considered as a promising alternative to traditional crops in a climate change scenario.

Quinoa in the Andean region grows from September to May, during the austral spring–summer time, with mean temperatures between 10 and 25°C. However, the most adequate range of mean temperature for its growth is 15–20°C (García *et al.*, 2015). The quinoa life cycle is approximately 6 months, but it varies depending on the region, which determines the sowing and harvest months (Sajjad *et al.*, 2014). Quinoa genotypes perform differently according to their latitude and altitude of origin (Bertero *et al.*, 1999*a*; Bhargava *et al.*, 2007; Curti *et al.*, 2016), showing important phenological differences in the duration of the phenological stages necessary to complete their life cycle and differing in their morphology at the canopy and inflorescence levels.

From a botanical standpoint, quinoa is a dicotyledonous plant approximately 0.3–3 m in height (Jacobsen & Stølen, 1993). The number of lateral branches it produces depends on its genotype and crop conditions. It has polymorphic leaves with serrate borders; usually, the basal leaves are larger with a rhomboidal or triangular shape, while leaves closer to the top are smaller and lanceolate. The lamina is covered by cells with granules of calcium oxalate, which allows water retention, increases in the relative humidity closer to the stomata and lowers evapotranspiration (Tapia *et al.*, 1979). Its inflorescence is

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a panicle that could present green, purple, red or a mix of red and purple colour. At physiological maturity the inflorescence colour can be green, white, purple, red, pink, yellow, orange, brown, grey, black or a combination such as red and white. The panicle shape can be rounded as a glomerular shape, elongated as an amaranthiform shape or have an intermediate shape between round and elongated. The panicle can be loose or compact when the pedicels are long or short, respectively (Tapia *et al.*, 1979; Biodiversity International & FAO, 2013). The flowers are small, without petals, perfect or pistillated with two to three feathery stigmas and five stamens. The flowering window lasts between 12 and 15 days, in which each flower remains open for 5–7 days (Tapia *et al.*, 1979). The fruit is an indehiscent achene that contains saponins in its outer layer (Tapia *et al.*, 1979; Rojas, 2003).

Although defining phenology is important in developing consistent information to support agronomical research and breeding programmes, there is still no international consensus on how to measure it in quinoa. In this crop, there is still a need to establish a standard scale as a unique criterion to quantify phenology, which could serve as a tool for researchers and plant breeders. Attempts to define the phenological stages in this crop were proposed in recent years by Mujica & Canahua (1989), Jacobsen & Stølen (1993), Ruiz & Bertero (2008) and Curti *et al.* (2016) (Table 1). All of these phenological scales were good contributions to the understanding of the crop although they have their weaknesses. Mujica & Canahua (1989) performed a detailed morphological scale but did not describe whether flowering was related to the primary or secondary stems of the plant, crucial in defining the beginning of the flowering window. Jacobsen & Stølen (1993) thoroughly described the reproductive stages but did not mention the vegetative growth stages of the plant and they did not include any figures of the important growth stages. Other weak points of this scale were the absence of descriptions of the phenological stages after anthesis, which includes the stages between grain filling and plant death. Ruiz & Bertero (2008) included in their proposed scale the phenological stages after anthesis. However, they missed a detailed description of each growth stage and omitted botanical illustrations. The scale by Curti *et al.* (2016) also did not include the stages after grain filling. In quinoa, none of the scales discussed above have been extensively used, possibly because of their current limitations. It is worth mentioning that the work of Bertero *et al.* (1996) described meticulously the changes in apical morphology during quinoa development and the results of Bertero *et al.* (1999*a*,*b*) recognised the vegetative and reproductive quinoa growth stages. These two last scales were not included in the table because they measured mainly apex development, which is a problem because phenological stages are meant to be easily recognised in the field, based on morphological characteristics without specialised equipment (Zadoks *et al.*, 1974).

From an international perspective, the Biologische Bundesanstalt Bundessortenamt und Chemische Industrie (BBCH) proposed a scale system to define the phenological events of agriculturally important plants and to fulfil the need of basic knowledge about the biology and the critical points of the plant's life cycle (Bleiholder *et al.*, 1989; Hack *et al.*, 1992). The BBCH scale consists of a decimal coding, dividing a plant's growth into a principal or general stage (0–9) and a secondary or specific stage (0–9), following the system proposed by Zadoks *et al.* (1974). Phenological scales using this widespread methodology have been described for plant species such as cereals (Meier *et al.*, 2009*a*), ornamentals (Meier *et al.*, 2009*b*), vegetables (Archontoulis *et al.*, 2010), fruit trees (Salazar *et al.*, 2006) and medicinals (Martinelli *et al.*, 2015). The scales were based on morphological characteristics that are easy to identify allowing descriptions of the phenological stages, while size measurements were represented as percentages of the final size, according to the species or variety. To fill the current gap in quinoa, the objective of this work is to adapt a phenology scale for this crop according to the BBCH guidelines as a practical tool to identify quinoa life cycle in a standard and internationally accepted language as an easily identifiable tool for everyone.

Materials and methods

Two Chilean quinoa genotypes, Ancovinto (AN) and Chiloe (CH), were used for the phenology records. Both genotypes were obtained through single plant selection by Francisco Fuentes (personal communication), the parental material for AN and CH was selected among the genotypes grown in Colchane (19°16′S, 68°37′W) and Ancud (41°52′S, 73°50′W) localities, respectively. The genotypes were grown in a greenhouse located at the Pontificia Universidad Católica de Chile, Santiago, Chile (33°27′S, 70°40′W). The seeds were sown on 4 of April in experimental units consisting of 0.25 m^2 (50 L) pots with 40 plants each, distributed in a minicanopy. Pots were filled with a mixture of 60% perlite (Harbolite® A-6 Harbolite Chile Ltda., Santiago, Chile) and 40% peat (Kekkila® Sphagnum peat Kekkila group, Vantaa, Finland) with a base fertiliser N : P : K (15 : 5 : 24). The average temperature in the greenhouse during the day was 18 ± 4°C and 16 ± 2°C at night. Plants were free of pests and diseases.

Four replicates per contrasting genotype were used to record phenology, ensuring that the scale would be useful to apply in all genotypes with the purpose to adapt the

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Table 1 Comparison of phenological scales of quinoa (*Chenopodium quinoa*) proposed by different authors: (A) Mujica & Canahua (1989), (B) Jacobsen & Stølen (1993), (C) Ruiz & Bertero (2008) and (D) Curti *et al.* (2016)

BBCH Code System		Mujica & Canahua (1989)		Jacobsen & Stølen (1993)		Ruiz & Bertero (2008)		Curti *et al.* (2016)	
Code	Description	Code	Description	Code	Description	Code	Description	Code	Description
0	Germination	1	Emergence	–	–	Vg	Vegetative	Stage 1	Emergence, first pair of leaves visible to anthesis
1	Leaf development	2	Two true leaves	–	–				
		3	Four true leaves	–	–				
		4	Six true leaves	–	–				
2	Formation of side shoots	5	Formation of side shoots	–	–				
3	Stem elongation	–	–	–	–				
4	Development of harvestable vegetative parts	–	–	–	–				
5	Inflorescence emergence	6	Beginning of panicle development	1	Covered bud	R	Reproductive		
		7	Panicle development	2	Visible bud				
		–	–	3	Differentiated bud				
		–	–	4	Bud of 0.5 cm length				
		–	–	5	Bud of 1.5 cm length				
		–	–	6	Pyramidal shape bud				
		–	–	7	Pyramidal shape bud differentiation				
6	Flowering	8	Beginning of flowering	8	Anthesis	Fl	Flowering	Stage 2	Anthesis
		9	Flowering anthesis	9	50% flowering			–	–
		–	–	10	100% flowering			–	–
		–	–	11	Beginning of floral dehiscence			–	–
		–	–	12	Floral dehiscence of most flowers			–	–
		–	–	13	Appearance wilted anthers			–	–
		–	–	14	33% seed set			–	–
		–	–	15	50% seed set			–	–
		–	–	16	67% seed set			–	–
		–	–	17	100% seed set			–	–
7	Fruit development	10	Milky grain	–	–	Gf	Grain filling	–	–
		11	Thick grain	–	–				
8	Ripening/maturity fruit	–	–	18	Maturity, most of the plant is green	E-PM	Emergence-physiological maturity	–	–
		12	Physiological maturity	19	Most of the plant is yellow				
		–	–	20	Mature grains			–	–
9	Senescence	–	–	21	Senescence				

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Table 2 Description of the phenological growth stages of quinoa (*Chenopodium quinoa*) according to the extended BBCH scale

BBCH Code Two-digit	Description
Principal growth stage 0: germination	
00	Dry seed
01	Initiation of seed imbibition
03	Seed imbibition completed
05	Radicle emergence from seed
07	Emergence of hypocotyl
08	Hypocotyl with cotyledons growing towards soil surface
09	Emergence of cotyledons through soil
Principal growth stage 1: leaf development	
10	Cotyledons fully emerged
11	First pair of leaves visible
12	Second pair of leaves visible
1.	Coding continues with the same scheme
19	Nine pair of leaves visible. If required, coding can continue following the same scheme.
Principal growth stage 2: formation of side shoots	
20	Visible lateral buds or expanded leaves without lateral stems
21	One side shoot visible
22	Two side shoots visible
2.	Coding continues with the same scheme
29	Nine side shoots visible. If required, coding can continue following the same scheme.
Principal growth stage 3: stem elongation (omitted)	
Principal growth stage 4: development of harvestable vegetative parts (omitted)	
Principal growth stage 5: inflorescence emergence	
50	Inflorescence present but still enclosed by leaves
51	Leaves surrounding inflorescence separated, inflorescence is visible from above
59	Inflorescence visible, but all the flowers are still closed
Principal growth stage 6: flowering	
60	Beginning of anthesis: main inflorescence flowers with first extruded anthers
67	Early end of anthesis: main inflorescence flowers with first senesced anthers
69	Complete anthesis: main inflorescence flowers with senesced anthers
Principal growth stage 7: fruit development	
70	Fruit set: ovary thickening and first visible grains in the main stem
Principal growth stage 8: ripening	
81	Milky grain, easily crushed with fingernails, liquid content and green pericarp
85	Thick grain, easily crushed with fingernails, white pasty content, green, beige, red or black pericarp
89	Ripe grain, difficult to crush with fingernails, dry content, the grain has a beige, red or black colour on its outside. Ready to harvest.
Principal growth stage 9: senescence	
91	Only basal leaves are dry
93	Leaves of the first half portion of the plant, starting from the base, are dead
95	All leaves are dead; stem colour turns from yellow to brown
97	Plant dead and dry
99	Harvested product

scale for quinoa growth habit variability. Phenology was recorded twice a week at 10:00 h from the appearance of the cotyledons onwards on two plants per replicate. The proposed scaled was constructed based on the BBCH system (Meier *et al.*, 2009*a*).

BBCH scale for *C. quinoa*

The description of the 10 principal growth stages was based on the two-digit BBCH code (Table 2). Principal stages as stem elongation (stage 3) and development of harvestable vegetative parts (stage 4) were omitted because of the quinoa characteristics. Stem elongation (stage 3) occurs simultaneously with leaf development (stage 1), formation of side shoots (stage 2), inflorescence emergence (stage 5) and flowering (stage 6), making it impossible to measure on a regular basis. Stage 3 has been previously omitted in the proposed BBCH scale of other species such as *Camelina sativa* (Martinelli & Galasso, 2011) and *Gossypium hirsutum* (Munger *et al.*,

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Figure 1 Phenological growth stages of quinoa (*Chenopodium quinoa*) according to the BBCH scale

1998) because in both species, stem elongation occurs parallel to leaf development (stage 1). Development of harvestable vegetative parts (stage 4) was omitted because in quinoa usually only fruits are harvested. Similar examples can be found in the BBCH scales of *C. sativa* (Martinelli & Galasso, 2011), *G. hirsutum* (Munger *et al.*, 1998), *Theobroma* sp. (Niemenak *et al.*, 2010), *Olea europaea* (Sanz-Cortés *et al.*, 2002), *Diospyros kaki* (García-Carbonell *et al.*, 2002), *Vitis vinifera* (Lorenz *et al.*, 1995), *Annona cherimola* (Cautín & Agustí, 2005), *Actinidia deliciosa* (Salinero *et al.*, 2009), *Mangifera indica* (Hernández Delgado *et al.*, 2011) and *Litchi chinensis* (Wei *et al.*, 2013).

Growth stages of quinoa (*C. quinoa*) based on the BBCH scale

Principal growth stage 0: germination

Principal growth stage 0 describes seed germination until the cotyledons emerge from the soil surface (Table 2). Secondary stages such as the initiation of seed imbibition (stage 01) (Fig. 1), seed imbibition completed (stage 03) and radicle emergence from the seed (stage 05) were included in this stage. Germination was epigeal followed by the emergence of the hypocotyl (stage 07) and the hypocotyl with the cotyledons growing towards the surface (stage 08). The germination stage finishes with the emergence of the cotyledons through the soil surface (stage 09) (Fig. 1).

Principal growth stage 1: leaf development

Principal growth stage 1 describes the emergence of the photosynthetic leaves on the main shoot (Table 2). Leaves emerged in pairs in quinoa, so a leaf pair was considered visible when the two leaf blades were separated from each other. This stage began when cotyledons were separated between them (stage 10) while the appearance of the photosynthetic leaves in the main shoot determined the secondary growth stage. The first pair of photosynthetic leaves visible was coded as stage 11 (Fig. 1), followed by the stage 12 and successively (Fig. 1).

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Figure 2 Phenological growth stages of inflorescence emergence (50), full flowering (60a) and thick grain at the main inflorescence level (85a) of quinoa (*Chenopodium quinoa*) according to the BBCH scale. Detailed flower (60b) and thick grain (85b) were included

Principal growth stage 2: formation of side shoots (secondary stems)

Principal growth stage 2 describes the formation of side shoots (Table 2) (Fig. 1). In quinoa, the appearance of side shoots can start before or after the inflorescence emergence, depending on genotype. For example, in this study, it was seen that CH genotype developed side shoots before the inflorescence emergence while AN generated side shoots after inflorescence emergence. For this reason, side shoot formation can be included together with the following principal growth stages using a diagonal stroke (example 23/65). A side shoot was considered visible when it showed one cm or more of length. The first side shoot was coded as stage 21, followed by the second side shoot coded as stage 22 and thereafter.

Principal growth stage 5: inflorescence emergence

Principal growth stage 5 qualitatively describes the development of the inflorescence in the main shoot (Table 2). Inflorescence emergence occurred before the end of the leaf development stage at the main shoot (stage 1). At first, the inflorescence buds were covered by young leaves and were not visible (stage 50) (Figs 1 and 2). After the leaves' elongation, the inflorescence becomes visible from above (stage 51). This stage finishes when the inflorescence becomes exposed without leaves covering it, although all its flowers are still closed (stage 59).

Principal growth stage 6: flowering

Principal growth stage 6 describes the development of the flowers within the main inflorescence (Table 2). Flowering starts when the first anthers were extruded (stage 60) (Figs 1 and 2). Anthesis was followed by the early end of flowering, which started with the first senesced anthers of the main inflorescence flowers (stage 67). At this stage, the inflorescence colour changed depending on the genotype, in which perigone colour changed as flowering progressed. When all the visible anthers of the main inflorescence senesced anthesis was considered to be completed (stage 69).

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Principal growth stage 7: fruit development

Principal growth stage 7 describes the fruit development in the main stem (Table 2). Fruit development started with fruit set (stage 70), which was defined as ovary thickening and the presentation of the first visible grains. Fruit development in quinoa was visible, but it was unpractical to follow the BBCH coding system that suggests using the percentage of the final grain diameter as a measurement for fruit development under field conditions. Considering that the final diameter of the quinoa achene is approximately 2 mm, that measurement methodology was not feasible to describe the rest of the secondary stages.

Principal growth stage 8: ripening

Principal growth stage 8 describes the ripening of the grains (Table 2). During the ripening process, the water content in the achene varied, modifying its texture and changing the pericarp colour, which transitioned from green to beige, red or black. To measure ripening (stage 8), the criteria proposed by Mujica & Canahua (1989) and Zadoks *et al.* (1974) was used, in which the grain was considered milky (stage 81), thick (stage 85) and mature (stage 89) (Figs 1 and 2). The milky stage was characteristic of grains with green pericarp that were easily crushed with the fingernails liberating a white liquid content. Thick grains, on the contrary, have a white pasty content when crushed, and their pericarps were green, beige, red or black. Ripe grains were difficult or impossible to crush and had a dry content; their pericarps were beige, red or black depending of their genotypes, and these grains were ready to harvest.

Principal growth stage 9: senescence

Principal growth stage 9 describes the plant senescence after fruit ripening (Table 2). Senescence started in the basal leaves (stage 91) (Fig. 1) and continued upwards, but the stem remained green (stage 93). Then, the remaining leaves became dead, and the stem turned from yellow to brown (stage 95). Finally, the entire plant was dead and dry (stage 97), and the product was then harvested (stage 99) (Fig. 1).

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